Exploring DeepFakes: Face Reenactment

Comparative Architecture Analysis: Autoencoders, GANs, FOMM & Rectified Flow

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Outline

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3. Method 2: GANs & FOMM
4. Method 3: Rectified Flow Models
5. Comparative Analysis
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Problem Formulation: Face Reenactment

Core Objective

- Animate static **Source (S)** photo using motion from **Driving (D)** video.
- Preserve identity of S, transfer motion dynamics from D.

Key Technical Challenges

- **Identity preservation:** Avoid warping face features into driving face.
- **Non-rigid deformation:** Facial skin stretches (not just rigid 2D shift).
- **Occlusion handling:** Mouth opening reveals unseen areas (teeth, tongue).

Input / Output Setup

Inputs:

- One static portrait photo (S)
- A driving video sequence (D)

Output:

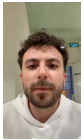
- Reenacted video of S
- Mimicking expressions and head pose of D

Dataset Preparation & Autoencoder Architecture

- **Goal & Motivation:** Swap faces between Person A and Person B while keeping motion. Built custom autoencoder to experience basics before pretrained tools.
- **Preprocessing:** Recorded short videos, extracted every 5th frame. Haar Cascade detection with **+25% margin**. Resized to 128×128 (0–1 scaled).
- **Architecture:** 1 shared Encoder (4 conv layers) and 2 Decoders (4 transposed-conv layers each, Sigmoid activation).
- **Face Swap Scheme:** Pass Person A through Decoder B, and Person B through Decoder A.

Subject	Prepared Faces	Resolution
Person A	100	128×128
Person B	138	128×128
Total	**238**	**128×128**

Input Videos (First Frames)



Person A (100 imgs)
[person_A.mp4]



Person B (138 imgs)
[person_B.mp4]

Custom Autoencoder: Training Loss & Video Outputs

Training Schedule (150 Epochs)

- First trained 50 epochs, saved weights.
- Resumed for another 100 epochs (150 total).

Epoch	Person A MSE	Person B MSE
146	0.003046	0.001939
147	0.003013	0.001929
148	0.003014	0.001905
149	0.002996	0.001882
150	**0.002935**	**0.001866**

Both models converged steadily; Person B achieved lower training loss.

Generated Video Outputs

Epochs	A → B Video	B → A Video
50 ep	`deepfake_A_to_B`	`deepfake_B_to_A`
150 ep	`deepfake_150_A_to_B`	`deepfake_150_B_to_A`



Comparison A → B (Input | 50ep | 150ep)

[Video: deepfake_150_A_to_B.mp4] [Video: comparison_A_to_B.mp4]

Custom Autoencoder: Observations & Failure Analysis

Why Did Face Swapping Fail?

- **No Visual Progress:** Comparing 50 vs. 150 epochs showed no discernible improvement.
- **Reconstruction vs. Identity:** The A $\rightarrow$ B face did not look like Person B, and B $\rightarrow$ A did not look like Person A.
- **Objective Flaw:** Pixel-wise MSE trained the model to recreate the original face, not transfer identity.
- **Dataset Size:** 238 total images were insufficient to learn a generalized, disentangled latent space.

Triplet Video Comparison Frame



Comparison B $\rightarrow$ A (Input | 50ep | 150ep)

[comparison_B_to_A.mp4 Video Link]

Key Takeaway

Shared latent code entangled identity with facial pose and expression.

Benchmarking Pretrained Models (FaceFusion)

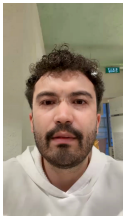
Inference Setup

- Tested 3 pretrained models in FaceFusion (no retraining).
- Source face gives identity; target video gives movement.

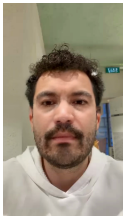
Model	Output Size	Frames	Duration
HyperSwap 1A	576×1024	576	$\sim$19.25 s
InSwapper 128	576×1024	576	$\sim$19.25 s
SimSwap 256	576×1024	576	$\sim$19.25 s

Visual Comparison

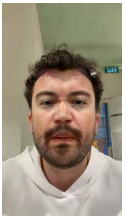
- **HyperSwap & InSwapper:** Fairly natural and consistent.
- **SimSwap:** Followed motion, but showed artifacts around eyes and skin texture.



HyperSwap 1A
[Video Link]



InSwapper 128
[Video Link]



SimSwap 256
[Video Link]

First frames: Erdem on Yusuf swaps

Autoencoders: Evaluation & Lessons Learned

Overall Evaluation

- Training our own autoencoder was not successful for face swapping.
- FaceFusion pretrained models gave much more promising results.
- Showed the importance of using models designed specifically for face swapping.
- Having only 100 images for Person A and 138 for Person B was a major limitation.

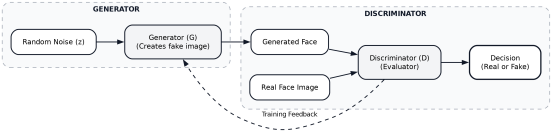
Lessons Learned (Erdem's Takeaways)

- **Low loss $\neq$ usable result:** The model got better at reconstructing faces, but not at swapping them. Another 100 epochs did not fix this.
- **Beyond running tools:** Building our own model helped us understand preprocessing, training, and evaluation hands-on.
- Pretrained models helped us compare how task-specific designs improve the result.

Standard GAN Architecture

How GANs Work (Goodfellow et al., 2014)

- Two neural networks competing in an adversarial setup:
- **Generator (G):** Takes noise **z** and learns to generate realistic face images.
- **Discriminator (D):** Evaluates images and decides if real or fake.
- **Adversarial training:** G gets feedback from D and learns photorealistic textures.



The Controllability Issue in Vanilla GANs

Why Unconditional GANs Fall Short

- Generates arbitrary faces from noise $\mathbf{z}$.
- Identity, pose, and emotion are entangled in $\mathbf{z}$.
- Cannot say: *"keep Person A, copy Person B's expression"*.

What Face Reenactment Requires

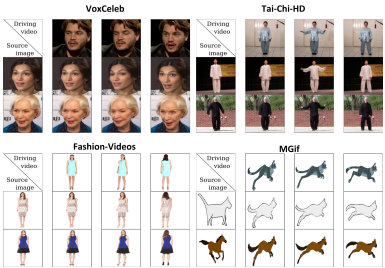
- Keep source identity fixed: appearance from S.
- Motion driven externally: dynamics from D.



First Order Motion Model (Siarohin et al., NeurIPS 2019)

Core Innovations

- **No Pretrained Landmarks:** Doesn't need 68 landmark detectors or 3D models (3DMM).
- **Self-Supervised:** Discovers keypoints automatically from raw video pairs.
- **Decoupled Repr.:** Appearance from source S; motion from driving D.
- **Domain Agnostic:** Works on human faces, bodies (Tai-Chi), and animals.

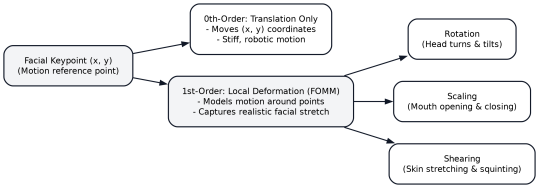


FOMM teaser: VoxCeleb, Tai-Chi, Fashion (Siarohin et al., 2019).

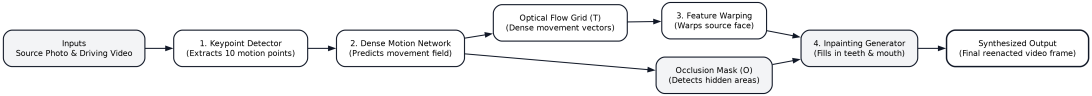
Motion Representation: 0th-Order vs. 1st-Order Motion

How Motion is Represented

- **0th-Order (Translation Only):**
  - Moves points in 2D space (x, y).
  - Cannot stretch skin; movement looks stiff and robotic.
- **1st-Order (FOMM / Local Affine):**
  - Estimates a Jacobian matrix around each point.
  - Captures rotation, scale (open mouth), and shear (skin stretch).



FOMM Architectural Pipeline



1. Keypoint Detector

Hourglass network extracts $K = 10$ unsupervised keypoints and Jacobians.

2. Dense Motion Network

Combines sparse keypoint motions into dense optical flow (T) and mask (O).

3. Inpainting Generator

Warps source features using flow T and synthesizes occluded areas ($O \approx 0$).

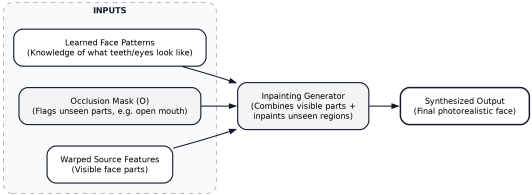
Self-Occlusion & Inpainting Mechanism

The Occlusion Problem

- Turning head or opening mouth reveals unseen areas in S (teeth, tongue).
- Pure warping has no source pixels $\rightarrow$ causes stretching or black holes.

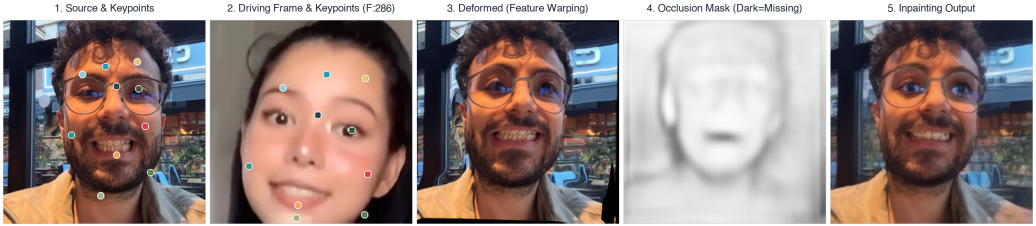
The FOMM Solution

- **Occlusion Mask (O):** Flags missing pixels ($O \approx 0$).
- **Inpainting:** Generator fills missing pixels using learned facial priors.



Pipeline Verification: Intermediate Stages

5 Key Stages Extracted from Our Local Inference Run



1. Source Points
S reference keypoints

2. Driving Points
D tracking points

3. Warped Frame
Flow-deformed source

4. Occlusion Mask
Dark = occluded regions

5. Inpainted Result
Final synthesized frame

Inference Pipeline: Python Pseudocode

Frame-by-Frame Execution Loop

```
# 1. Extract reference keypoints from source portrait
kp_source = keypoint_detector(source_image)

# 2. Iterate through driving video frames
for frame_d in driving_video:
    # Step 1: Motion tracking
    kp_driving = keypoint_detector(frame_d)

    # Step 2: Dense motion & occlusion estimation
    flow, occ = dense_motion_net(kp_source, kp_driving)

    # Step 3: Flow warping & inpainting
    out_frame = generator(source_image, flow, occ)

    yield out_frame
```

Core Modules

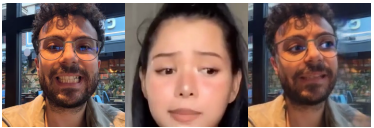
- `keypoint_detector`: Extracts 10 unsupervised keypoints.
- `dense_motion_net`: Computes optical flow and occlusion mask.
- `generator`: Warps source pixels and inpaints mouth/teeth.

Architectural Property

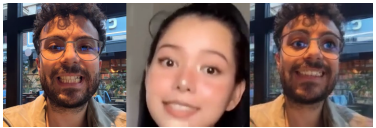
No Recurrence: The loop carries no hidden state across frames ($t \rightarrow t+1$).

Experimental Results: Bella Poarch Benchmark

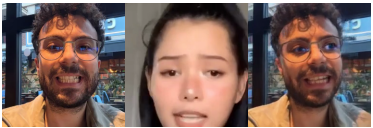
Triplets: [Source Portrait | Driving Frame | Synthesized Result]



Frame 30 (1.0s): Head Tilt



Frame 286 (9.5s): Smile & Squint



Frame 20 (0.7s): Mouth Opening

Demo Video Link

- **3×3 Multi-Subject Comparison Video (All Members):** Watch Online (External Link)

Failure Mode Experiment: Dental Braces

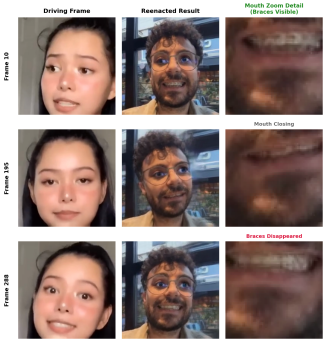
Experimental Setup

- **Goal:** Test FOMM on rare features absent from training data (OOD).
- **Subject:** Source portrait with metallic dental braces.
- **Motion:** Sequence from head tilt to lip closure and wide smile.

Hypothesis & Expected Regimes

- VoxCeleb training data has no faces with braces.
- **Warping ($O \approx 1$):** Samples source pixels $\rightarrow$ braces preserved.
- **Inpainting ($O \approx 0$):** Large deformation masks mouth $\rightarrow$ model synthesizes generic teeth.
- **Prediction:** Inpainting erases braces using dataset average.

Failure Mode: Vanishing Dental Braces



Frame 10 (top), Frame 195 (middle), Frame 288 (bottom).

Observed Generation Regimes:

- **Frame 10 (0.3s) — Warping ($O \approx 1$):** Small shift preserves flow; metallic brackets transfer cleanly.
- **Frame 195 (6.5s) — Lip Closure:** Lips close, naturally hiding mouth interior.
- **Frame 288 (9.6s) — Inpainting ($O \approx 0$):** Extreme smile triggers inpainting; generator paints generic teeth, erasing braces.

Technical Mechanism: Dual-Path Synthesis

Synthesis Formulation

$$\text{Output} = (\text{Warped } S \odot O) + (G_{\text{inpaint}} \odot (1 - O))$$

Operational Pathways

- **Warping ($O \approx 1$):** Copies from source S, preserving identity details (braces).
- **Inpainting ($O \approx 0$):** Under deformation, synthesizes pixels from VoxCeleb average (plain teeth).

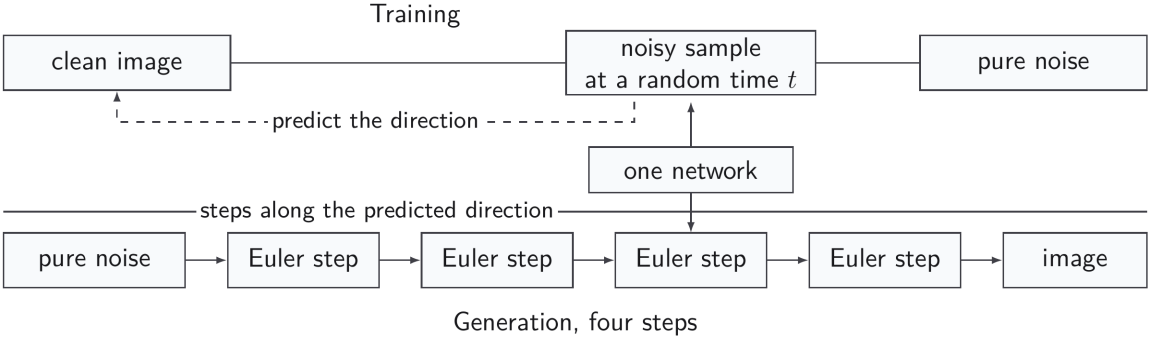
Core Finding

- Pixel fidelity depends on the active pathway.
- $O \approx 1$ regions preserve source specifics.
- $O \approx 0$ regions reflect training dataset bias.

Failure Mode Demo Video

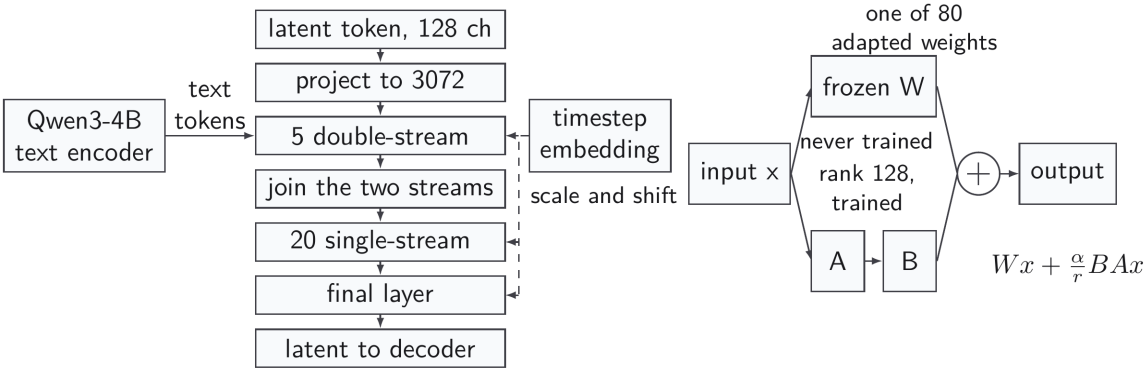
- **Dental Braces Video:**
  Watch Online (External Link)

Rectified Flow Fundamentals



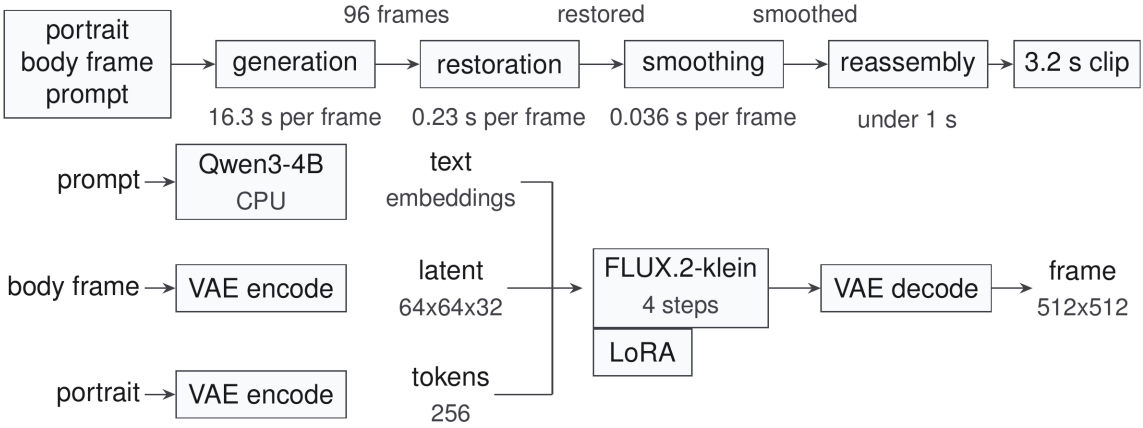
- **Training.** The image and noise mix on one straight line at a random time.
- **Generation.** The model predicts the direction, and the sampler steps along it.
- **Our setup.** A released checkpoint, already step distilled, so four steps are enough.

Architecture: FLUX.2-klein and LoRA



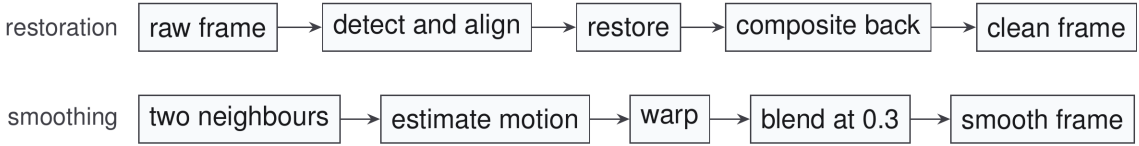
- **Flow model.** A 4 billion parameter rectified flow transformer.
- **Layers.** Five double-stream blocks, then 20 single-stream blocks.
- **Adapter.** It adapts 80 weights, and the base weights stay frozen.

The Generation Pipeline on 4 GB



- **Why it fits.** The text encoder runs once on the CPU, then its memory is freed.
- **Safety gate.** A frame with more than one face is refused, and there is no bypass.

Cleanup Stages and Results



Stage	Face change	Consecutive identity	Identity to portrait	Hair noise
Control	6.657	0.985	−0.036	31.600
Raw swap	7.451	0.923	0.215	32.570
Restoration	7.160	0.927	0.215	32.072
Flow on raw	6.666	0.939	0.215	32.060
Restoration + flow	6.630	0.938	0.215	31.731
Deflicker	5.627	0.936	0.211	31.896

- **Recommended.** Restoration, then optical flow at strength 0.3.
- **Rejected.** The per-pixel deflicker is cheaper, but it removes real motion too.

Reassembly and the Boil



Raw assembly



Flow on the raw frames



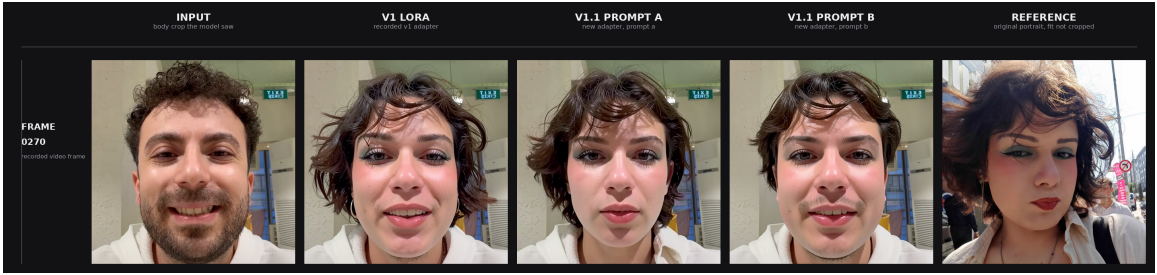
Restoration + flow

- **Reassembly.** H.264 at 29.969 fps, with the original audio muxed back.
- **Watch the boil.** It is clear in the first clip and gone in the last one.

Adapter Versions and Failure Analysis

	v1	v1.1
Seconds per frame	16.34	about 199
Peak VRAM	3.60 GiB	**4.68 GiB**
Adapter rank	128	**512**
96 frames	26 min	about 5.3 h

- **Measured.** Only v1 ran the metric script.
- **Memory.** v1.1 went past the card, 12x slower.
- **Judgement.** We preferred v1 by eye, so the clip kept v1.
- **Tail.** The worst frames still jump.



Five columns: input body crop, v1, v1.1 with prompt A, v1.1 with prompt B, and the reference portrait.

Comparative Analysis

Autoencoders vs. GANs (FOMM) vs. Rectified Flow Models

[Placeholder – Team Synthesis]

Ethical Implications & DeepFake Detection

Societal Risks

- **Impersonation:** Reenacting individuals without their consent.
- **Misinformation:** Generating fabricated videos to deceive the public.

Detection Strategies

- **Biological signals:** Unnatural blinking and pulse signals (rPPG).
- **Frequency artifacts:** High-frequency noise from neural upsampling.
- **Watermarking / C2PA:** Cryptographic metadata embedded at capture.

Conclusion

Summary

- Animates portraits using 10 unsupervised keypoints and Jacobians.
- Eliminates need for landmark detectors or 3D face models.
- Inpainting handles disocclusion, but rare features (braces) vanish due to dataset bias.

Project Repository

Code, inputs, and outputs:

`https://github.com/yusufarslan/face-reenactment`

Questions & Discussion